Information and Network Security

Practical Journal

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Practical 1

1] Caesar Cipher

```
import string

LETTERS = string.ascii_uppercase
PT = "HELLOWORLD"
key = 3
CT = ""
for i in PT:
     p = LETTERS.rfind(i)
     CT += LETTERS[(p+key)%26]

print("Plain text:",PT,"\nCipher text:",CT)
```

```
----------------------
Plain text: HELLOWORLD
Cipher text: KHOORZRUOG
```

2] Monoalphabetic Cipher

```
import string
import random

LETTERS = string.ascii_uppercase
PT = "HELLOWORLD"
CT = ""
i = 0
while i < len(PT):
     n = random.randint(0,25)
     if (LETTERS.rfind(PT[i]) != n):
          CT += LETTERS[n]
     i += 1

print("Plain text:",PT,"\nCipher text:",CT)
```

```
======================
Plain text: HELLOWORLD
Cipher text: FFSOPONXTA
```

Practical 2

Vernam Cipher

```
import random
import string

LETTERS = string.ascii_uppercase
PT = "HELLOWORLD"

OTP = ""
CT = ""

for _ in range(len(PT)):
     OTP += random.choice(LETTERS)

print("OTP:",OTP)

for i in range(len(PT)):
     m1 = LETTERS.rfind(PT[i])
     m2 = LETTERS.rfind(OTP[i])
     s = m1 + m2
     #print(m1, m2, s)

     if s>= 26:
          CT += LETTERS[s-26]
     else:
          CT += LETTERS[s]

print("Plain text:", PT, "\nCipher text:", CT)
```

```
----------------------
OTP: OHXTQDBASB
Plain text: HELLOWORLD
Cipher text: VLIEEZPRDE
```

Practical 3

Diffie-Hellman

```
n = int(input("Enter prime no. 1:"))
g = int(input("Enter prime no. 2:"))
x = int(input("Enter A's key:"))
y = int(input("Enter B's key:"))

A = g ** x % n
B = g ** y % n

K1 = B ** x % n
K2 = A ** y % n

print(f"A:{A}\nB:{B}\nK1:{K1}\nK2:{K2}")

if K1 == K2:
     print("Keys transmitted successfully!")
else:
     print("Keys don't match")
```

```
Enter prime no. 1:29
Enter prime no. 2:23
Enter A's key:45
Enter B's key:68
A:16
B:25
K1:23
K2:23
Keys transmitted successfully!
```

Practical 4

DES Algorithm

Install package: `pip install pycrytodome`

```
from Crypto.Cipher import DES
from Crypto.Util.Padding import pad, unpad

key = b'hello123'
txt= b'asd'

des = DES.new(key, DES.MODE_ECB)

padded_txt = pad(txt, DES.block_size)
print(f"Padded text:{padded_txt}")

encrypted_txt = des.encrypt(padded_txt)
print(f"Encrypted text:{encrypted_txt}")

decrypted_padded_txt = des.decrypt(encrypted_txt)
decrypted_txt = unpad(decrypted_padded_txt, DES.block_size)

print(f"Decrypted text:{decrypted_txt}")
```

```
---------------------- RESTART: D:/
Padded text:b'asdz\x04\x04\x04\x04'
Encrypted text:b'W.t">\x92\x93\x91'
Decrypted text:b'asdz'
```

Practical 5

AES Algorithm

Install package: `pip install pycrytodome`

```
from Crypto.Cipher import AES

key = b'Sixteen byte key'
data = b'hello from nether'

e_cipher = AES.new(key, AES.MODE_EAX)
e_data = e_cipher.encrypt(data)

d_cipher = AES.new(key, AES.MODE_EAX, e_cipher.nonce)
d_data = d_cipher.decrypt(e_data)

print(f"Encrypted data:{e_data}")
print(f"Decrypted data:{d_data}")
```

```
Encrypted data:b'\xecx\x0e\x1e]\x8b\xd0\xf8M\xb8O0\xdf#\xe6\xe5\xac'
Decrypted data:b'hello from nether'
```

Practical 6

RSA Algorithm

Install package: `pip install rsa`

```
import rsa

(bob_pub, bob_priv) = rsa.newkeys(512)
message = "hello world"
print("original message:",message)
message = message.encode('utf8')
print()

encrypted = rsa.encrypt(message,bob_pub)
print("encrypted message")
print(encrypted)
print()

decrypted = rsa.decrypt(encrypted, bob_priv)
print("decrypted message:",decrypted)
```

```
original message: hello world

encrypted message
b'{\xd5)\x18b\x07\xfdn\xe4\x14>\x0b\xb8!\xe2\xee\xf5z\xec
hp\xef\x7f+\xc7\x82\x8c\xe9\xe4U\x14\xebP\x00G\x01\xd2S\x
05\xfb\x99\xa7%\xd7>\xf1#\xa9\x13\xbe](1s\xdf}n<\xee\xee
\xa6\xbc\x88'

decrypted message: b'hello world'
```

Practical 7

Message Digest

```
import hashlib

hash_obj = hashlib.md5(b'Hello, Python!')
print(hash_obj.hexdigest())

hash_obj = hashlib.md5(b'Hello, Python!')
print(hash_obj.digest())

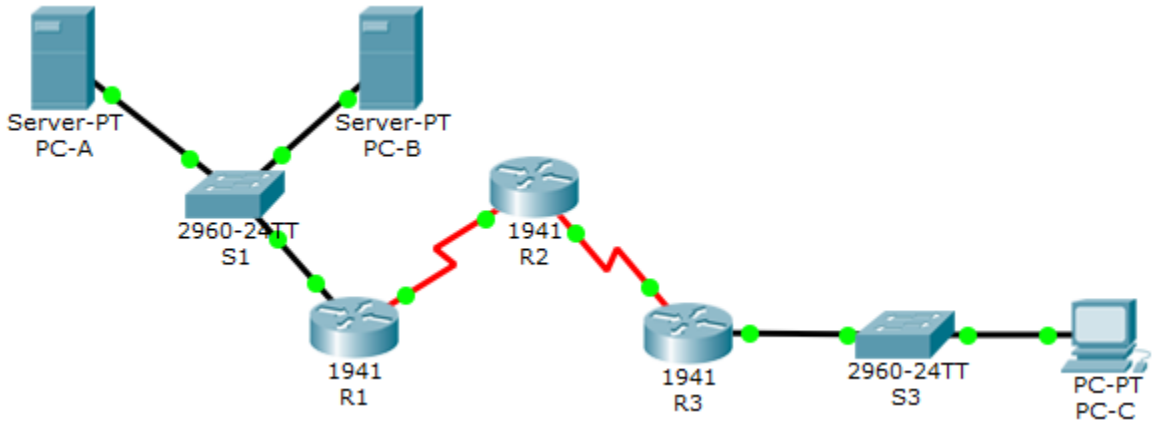
mystring = input("Enter string to hash:")
hash_obj = hashlib.md5(mystring.encode())
print(hash_obj.hexdigest())
```

```
a0af7810eb5fcb84c730f851361de06a
b'\xa0\xafx\x10\xeb_\xcb\x84\xc70\xf8Q6\x1d\xe0j'
Enter string to hash:sdlgasgd
dc9e6600721bde812fa828c7d72b7f61
```

Practical 8

Configure Cisco Routers for Syslog, NTP, and SSH Operations

Topology:



Addressing Table:

Device	Interface IP	Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.1	255.255.255.252	N/A	N/A
PC-A	NIC	192.168.1.5	255.255.255.0	192.168.1.1	S1 F0/6
PC-B	NIC	192.168.1.6	255.255.255.0	192.168.1.1	S2 F0/18
PC-C	NIC	192.168.3.5	255.255.255.0	192.168.3.1	S3 F0/18

Note: Add switch port to the routers.

- **Router OSPF CLI Commands:**

R1:

```
Router#en
Router#conf t
Enter configuration commands, one per line.  End with CNTL,
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 10.1.1.0 0.0.0.3 area 0
Router(config-router)#show ip ospf interface
```

R2:

```
Router#en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 10.1.1.0 0.0.0.3 area 0
Router(config-router)#network
00:18:07: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial0/1/0 from LOADING
to FULL, Loading Done
10.2.2.0 0.0.0.3 area 0
Router(config-router)#
```

R3:

```
Router#en
Router#conf t
Enter configuration commands, one per line.  End with CNTL,
Router(config)#router ospf 1
Router(config-router)#network 192.168.3.0 0.0.0.255 area 0
Router(config-router)#network 10.2.2.0 0.0.0.3 area 0
```

1. Configure OSPF MD5 Authentication

Step1: Configure OSPF MD5 authentication for all the routers:

```
Router(config)#router ospf 1
Router(config-router)#area 0 authentication message-digest
```

Step2: Configure the MD5 key for all the routers:

R1:

```
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip ospf mes
Router(config-if)#ip ospf message-digest-key 1 md5 MD5pa55

Router(config)#interface Serial0/1/0
Router(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
```

R2:

```
Router(config)#interface Serial0/1/0
Router(config-if)#ip ospf mes
Router(config-if)#ip ospf message-digest-key 1 md5 MD5pa55

Router(config)#interface Serial0/1/1
Router(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
```

R3:

```
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip ospf mes
Router(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
```

```
Router(config)#interface Serial0/1/1
Router(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
```

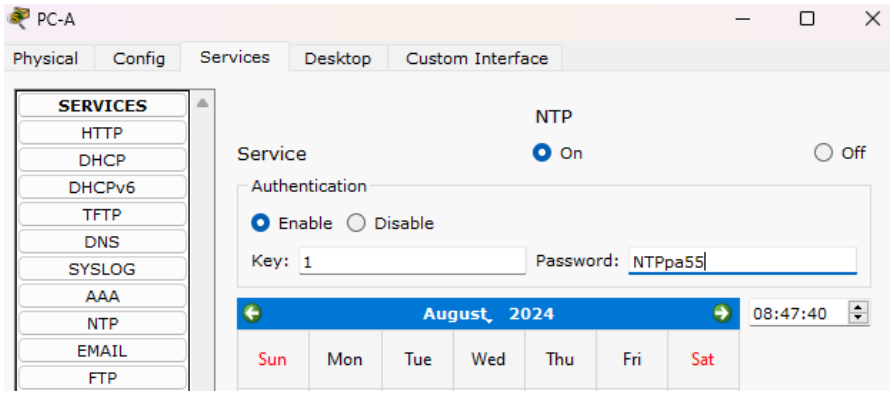
Step3: Verify the MD5 authentication configurations:

```
Router>show ip ospf interface

GigabitEthernet0/1 is up, line protocol is up
  Internet address is 192.168.1.1/24, Area 0
  Process ID 1, Router ID 192.168.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 192.168.1.1, Interface address 192.168.1.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:05
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    Youngest key id is 1
Serial0/1/0 is up, line protocol is up
  Internet address is 10.1.1.1/30, Area 0
  Process ID 1, Router ID 192.168.1.1, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT, Priority 0
  No designated router on this network
 --More--
```

2. Configure NTP

Step1: Enable NTP authentication on PC-A:



Step2: Configure R1, R2, and R3 as NTP clients:

```
R1(config)#ntp server 192.168.1.5
R2(config)#ntp server 192.168.1.5
R3(config)#ntp server 192.168.1.5
```

Verify client configuration using the command `show ntp status`:

```
R3#show ntp status
Clock is synchronized, stratum 2, reference is 192.168.1.5
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**19
reference time is EA45FA12.000001A1 (08:55:46.417 UTC Thu Aug 22 2024)
clock offset is 0.00 msec, root delay is 0.00  msec
root dispersion is 0.02 msec, peer dispersion is 0.02 msec.
```

Step3: Configure routers to update hardware clock:

```
R1(config)#ntp update-calendar
R2(config)#ntp update-calendar
R3(config)#ntp update-calendar
```

Verify that the hardware clock was updated using the command `show clock`:

```
R3#show clock
*8:59:15.630 UTC Thu Aug 22 2024
```

Step4: Configure NTP authentication on the routers:

```
R1(config)#ntp authenticate
R1(config)#ntp trusted-key 1
R1(config)#ntp authentication-key 1 md5 NTPpa55


R2(config)#ntp authenticate
R2(config)#ntp trusted-key 1
R2(config)#ntp authentication-key 1 md5 NTPpa55


R3(config)#ntp authenticate
R3(config)#ntp trusted-key 1
R3(config)#ntp authentication-key 1 md5 NTPpa55
```

Step5: Configure routers to timestamp log messages:

```
R1(config)# service timestamps log datetime msec
R2(config)# service timestamps log datetime msec
R3(config)# service timestamps log datetime msec
```

3. Configure Routers to Log Messages to the Syslog Server

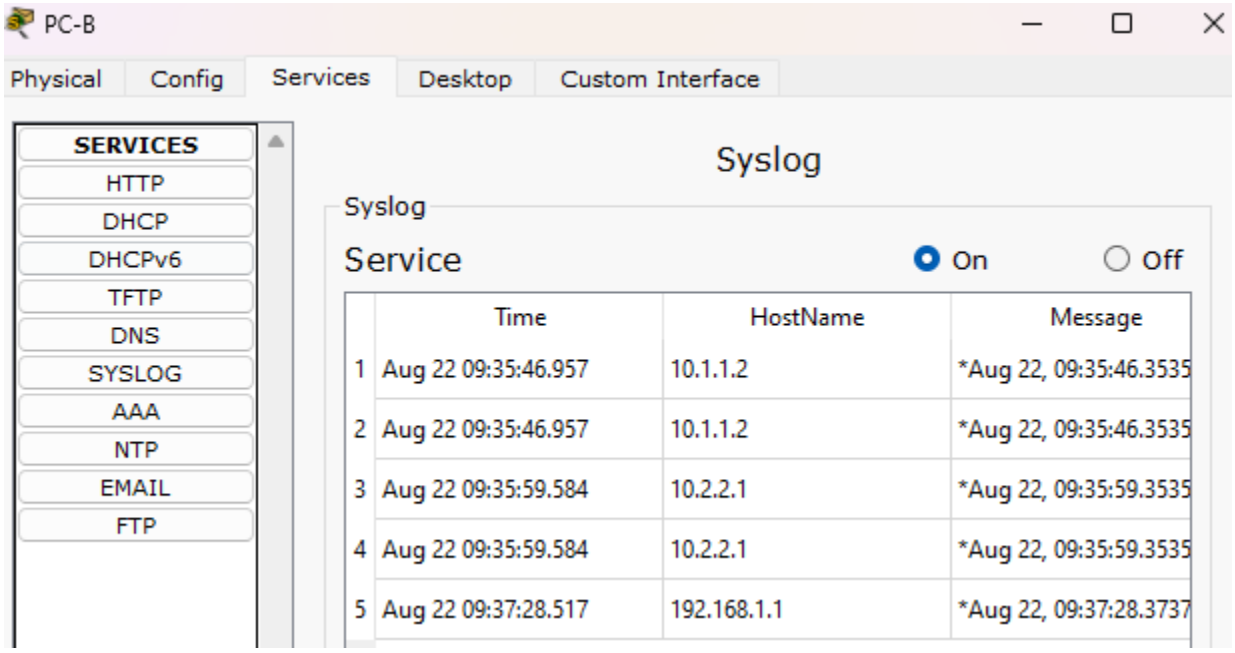
Step1: Configure the routers to identify the remote host (Syslog Server) that will receive logging messages.

```
R1(config)# logging host 192.168.1.6
R2(config)# logging host 192.168.1.6
R3(config)# logging host 192.168.1.6
```

Step2: Verify logging configuration using command `**show logging**`:

```
R3#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
          0 flushes, 0 overruns, xml disabled, filtering disabled)
```

Step3: Examine logs of the Syslog Server:



4. Configure R3 to Support SSH Connections

Step1: Configure a domain name:

```
R3(config)#ip domain-name ccnasecurity.com
```

Step2: Configure users for login to the SSH server on R3:

```
R3(config)#username SSHadmin privilege 15 secret
ciscosshpa55
```

Step3: Configure the incoming vty lines on R3:

```
R3(config)#line vty 0 4
R3(config-line)#login local
R3(config-line)#transport input ssh
```

Step4: Erase existing key pairs on R3:

```
R3(config-line)#crypto key zeroize rsa
```

Note: If no keys exist, you might receive this message:

```
% No Signature RSA Keys found in configuration.
```

Step5: Generate the RSA encryption key pair for R3:

```
R3(config)#crypto key generate rsa
The name for the keys will be: R3.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
```

Step6: Verify the SSH configuration using the command `**show ip ssh**`:

```
R3#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
```

Step7: Configure SSH timeouts and authentication parameters:

```
R3(config)#ip ssh time-out 90
R3(config)#ip ssh authentication-retries 2
R3(config)#ip ssh version 2
```

Step8: Attempt to connect to R3 via Telnet from PC-C:

```
PC>telnet 192.168.3.1
Trying 192.168.3.1 ...Open

[Connection to 192.168.3.1 closed by foreign host]
```

Step9: Connect to R3 using SSH on PC-C:

```
PC>ssh -l SSHadmin 192.168.3.1
Open
Password:



R3#
```

Step10: Connect to R3 using SSH on R2:

```
R2#ssh -v 2 -l SSHadmin 10.2.2.1
Open
Password:

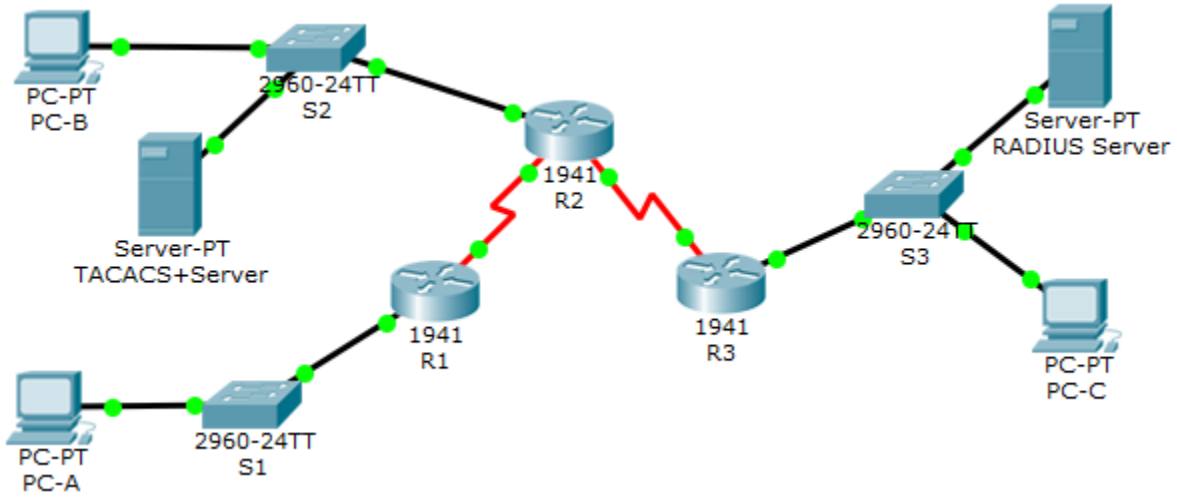


R3#
```

Practical 9

Configure AAA Authentication on Cisco Routers

Topology:



Addressing Table:

Device	Interface IP	Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/1
	S0/0/0 (DCE)	10.1.1.2	255.255.255.252	N/A	N/A
R2	G0/0	192.168.2.1	255.255.255.0	N/A	S2 F0/2
	S0/0/0	10.1.1.1	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.1	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.2	255.255.255.252	N/A	N/A
TACACS +Server	NIC	192.168.2.2	255.255.255.0	192.168.2.1	S2 F0/6
Radius Server	NIC	192.168.3.2	255.255.255.0	192.168.3.1	S3 F0/1
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/2
PC-B	NIC	192.168.2.3	255.255.255.0	192.168.2.1	S2 F0/1
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18

1. Configure Local AAA Authentication for Console Access on R1

Step1: Configure a local username on R1:

```
R1(config)#username Admin1 secret admin1pa55
```

Step2: Configure local AAA authentication for console access on R1:

```
R1(config)#aaa new-model
R1(config)#aaa authentication login default local
```

Step3: Configure the line console to use the defined AAA authentication method:

```
R1(config)#line console 0
R1(config-line)#login authentication default
```

Step4: Verify the AAA authentication method:

```
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#exit


Press RETURN to get started!



User Access Verification

Username: Admin1
Password:
R1>
```

2. Configure Local AAA Authentication for vty lines on R1

Step1: Configure domain name and crypto key for use with SSH:

```
R1(config)#ip domain-name ccnasecurity.com
R1(config)#crypto key generate rsa
The name for the keys will be: R1.ccnasecurity.com
Choose the size of the key modulus in the range of
  General Purpose Keys. Choosing a key modulus greai
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exp
```

Step2: Configure a named list AAA authentication method for the vty lines on R1:

```
R1(config)#aaa authentication login SSH-LOGIN local
*Mar 1 1:30:49.523:  %SSH-5-ENABLED: SSH 1.99 has been enabled
```

Step3: Configure the vty lines to use the defined AAA authentication method:

```
R1(config)# line vty 0 4
```

```
R1(config-line)#login authentication SSH-LOGIN
R1(config-line)#transport input ssh
```

Step4: Verify the AAA authentication method:

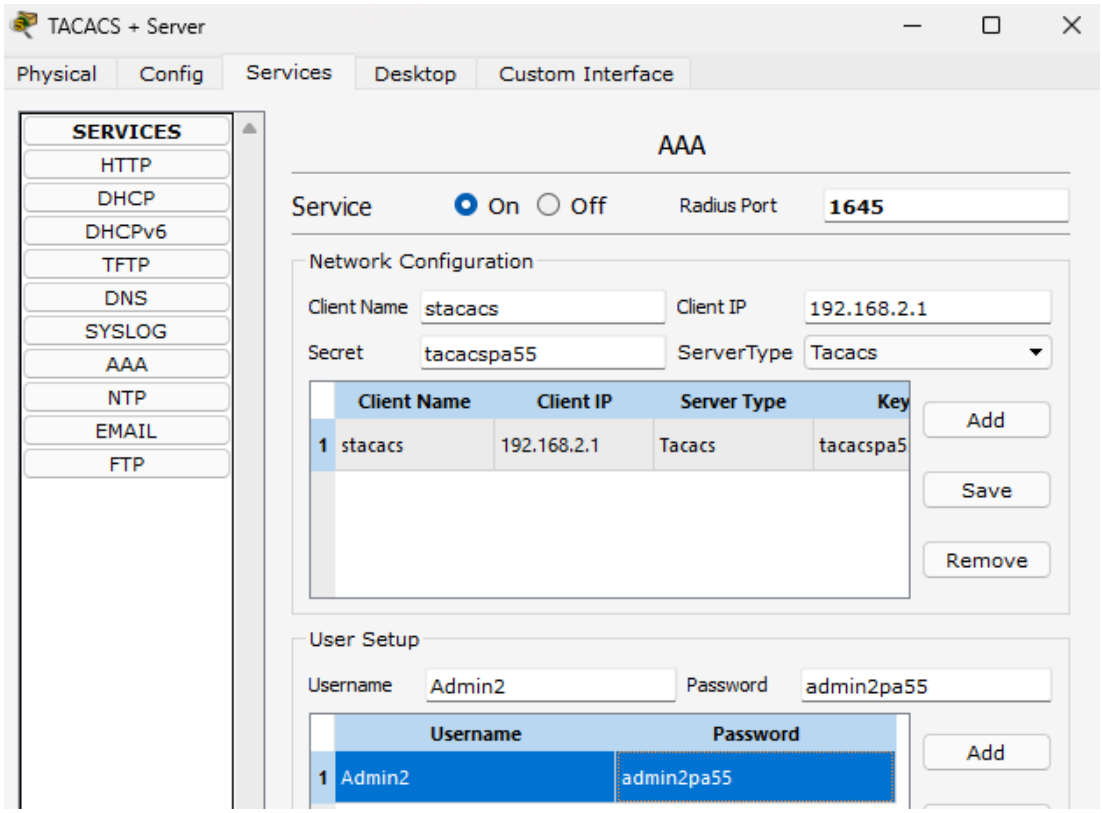
```
PC>ssh -l Admin1 192.168.1.1
Open
Password:
R1>
```

3. Configure Server-Based AAA Authentication Using TACACS+ on R2

Step1: Configure a backup local database entry called Admin:

```
R2(config)# username Admin2 secret admin2pa55
```

Step2: Verify the TACACS+ Server configuration:



Step3: Configure the TACACS+ server specifics on R2:

```
R2(config)#tacacs-server host 192.168.2.2
R2(config)#tacacs-server key tacacspa55
```

Step4: Configure AAA login authentication for console access on R2:

```
R2(config)#aaa new-model
```

```
R2(config)#aaa authentication login default group tacacs+
local
```

Step5: Configure the line console to use the defined AAA authentication method:

```
R2(config)#line console 0
R2(config-line)#login authentication default
```

Step6: Verify the AAA authentication method:

```
User Access Verification

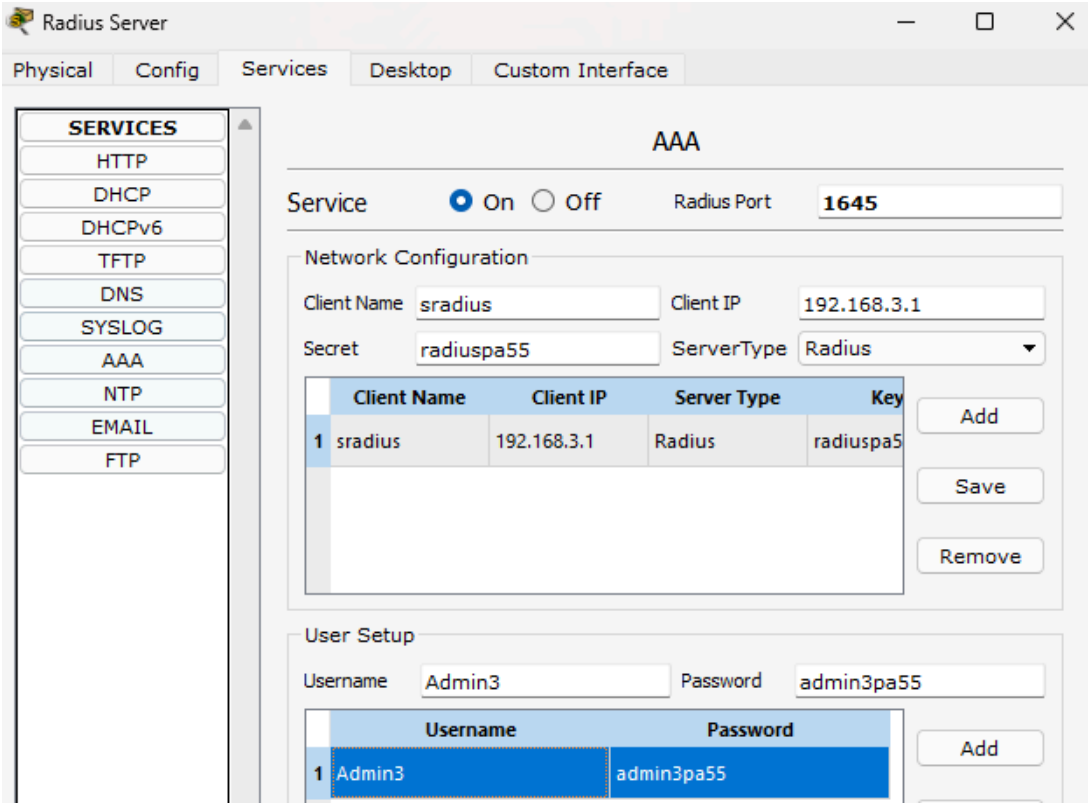
Username: Admin2
Password:
```

4. Configure Server-Based AAA Authentication Using RADIUS on R3

Step1: Configure a backup local database entry called Admin:

```
R3(config)# username Admin3 secret admin3pa55
```

Step2: Verify the RADIUS Server configuration:



Step3: Configure the RADIUS server specifics on R3:

```
R3(config)# radius-server host 192.168.3.2
R3(config)# radius-server key radiuspa55
```

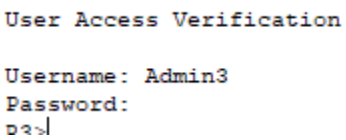
Step4: Configure AAA login authentication for console access on R2:

```
R3(config)#aaa new-model
R3(config)#aaa authentication login default group radius
local
```

Step5: Configure the line console to use the defined AAA authentication method:

```
R3(config)#line console 0
R3(config-line)#login authentication default
```

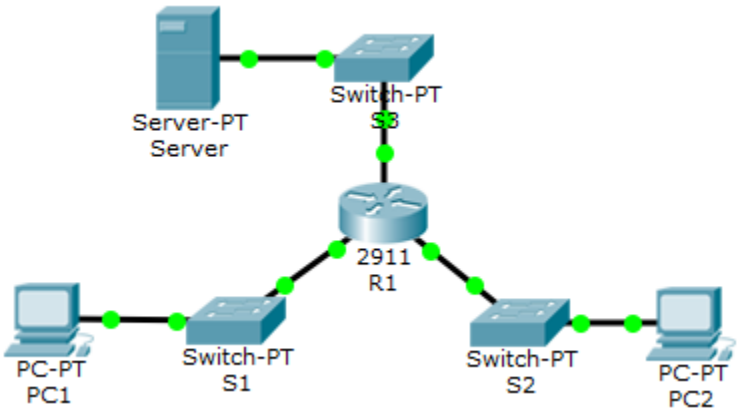
Step6: Verify the AAA authentication method:



Practical 10

Configuring Extended ACLs - Scenario 1

Topology:



Addressing Table:

Device	Interface IP	Address	Subnet Mask	Default Gateway
R1	G0/0	172.22.34.65	255.255.255.224	N/A
	G0/1	172.22.34.97	255.255.255.240	N/A
	G0/2	172.22.34.1	255.255.255.192	N/A
Server	NIC	172.22.34.62	255.255.255.192	172.22.34.1
PC1	NIC	172.22.34.66	255.255.255.224	172.22.34.65
PC2	NIC	172.22.34.98	255.255.255.240	172.22.34.97

1. Configure, Apply and Verify an Extended Numbered ACL

Step1: Configure an ACL to permit FTP and ICMP:

```
R1(config)# access-list 100 permit tcp 172.22.34.64
0.0.0.31 host 172.22.34.62 eq ftp
R1(config)# access-list 100 permit icmp 172.22.34.64
0.0.0.31 host 172.22.34.62
```

Step2: Apply the ACL on the correct interface to filter traffic:

```
R1(config)#interface GigabitEthernet0/0
R1(config-if)#ip access-group 100 in
```

Step3: Verify the ACL implementation:

a. Ping from PC1 to Server. If the pings are unsuccessful, verify the IP addresses before continuing.
b. FTP from PC1 to Server. The username and password are both `cisco`.
```
PC> ftp 172.22.34.62
```
c. Exit the FTP service of the Server.
```
ftp> quit
```
d. Ping from PC1 to PC2. The destination host should be unreachable, because the traffic was not explicitly permitted.

2. Configure, Apply and Verify an Extended Named ACL

Step1: Configure an ACL to permit HTTP access and ICMP:

- You can configure named standard and extended ACLs. This access list filters both source and destination IP addresses; therefore, it must be extended. Enter HTTP_ONLY as the name. (For Packet Tracer scoring, the name is case-sensitive.)
```
R1(config)# ip access-list extended HTTP_ONLY
```

- All devices on the PC2 LAN need TCP access. Enter the network address & wildcard, specify the server address and filter `www` traffic.
```
R1(config-ext-nacl)# permit tcp 172.22.34.96 0.0.0.15 host 172.22.34.62 eq www
```

- Create a second access list statement to permit ICMP (ping, etc.) traffic from PC2 to Server.
```
R1(config-ext-nacl)# permit icmp 172.22.34.96 0.0.0.15 host 172.22.34.62
```

- All other traffic is denied, by default. Exit out of extended named ACL configuration mode.

Step2: Apply the ACL on the correct interface to filter traffic:
```
R1(config)# interface gigabitEthernet 0/1
R1(config-if)# ip access-group HTTP_ONLY in
```

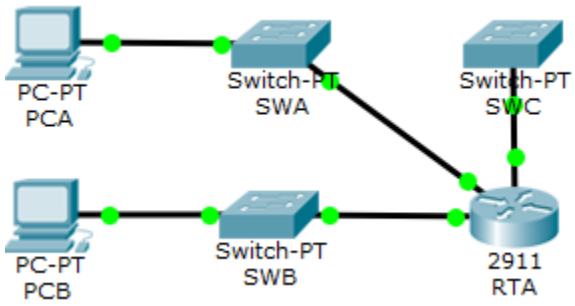
Step3: Verify the ACL implementation:

a. Ping from PC2 to Server. The ping should be successful, if the ping is unsuccessful, verify the IP addresses before continuing.
b. FTP from PC2 to Server. The connection should fail.
c. Open the web browser on PC2 and enter the IP address of Server as the URL. The connection should be successful.

Practical 11

Configuring Extended ACLs - Scenario 2

Topology:



Addressing Table:

Device	Interface IP	Address	Subnet Mask	Default Gateway
R1	G0/0	10.101.117.49	255.255.255.248	N/A
	G0/1	10.101.117.33	255.255.255.240	N/A
	G0/2	10.101.117.1	255.255.255.224	N/A
PCA	NIC	10.101.117.51	255.255.255.248	10.101.117.49
PCB	NIC	10.101.117.35	255.255.255.240	10.101.117.33
SWA	VLAN 1	10.101.117.50	255.255.255.248	10.101.117.49
SWB	VLAN 1	10.101.117.34	255.255.255.240	10.101.117.33
SWC	VLAN 1	10.101.117.2	255.255.255.224	10.101.117.1

For Switches:

```
SWA(config)# interface vlan 1
SWA(config-if)# ip address <ip addr> <subnet mask>
SWA(config-if)# no shutdown
SWA(config-if)# exit
SWA(config)# ip default-gateway <gateway ip>
```

Step1: Configure the extended ACL:

```
     Router(config)#access-list 199 permit tcp 10.101.117.32
0.0.0.15 10.101.117.0 0.0.0.31 eq 22
```

```
Router(config)#access-list 199 permit icmp any any
```

Step2: Apply the extended ACL:

```
Router(config)# interface gigabitEthernet 0/2
Router(config-if)# ip access-group 199 out
```

Step3: Verify the extended ACL implementation.

a. Ping from PCB to all of the other IP addresses in the network. If the pings are unsuccessful, verify the IP addresses before continuing.
b. SSH from PCB to SWC. The username is Admin, and the password is Adminpa55.

```
PC> ssh -l Admin 10.101.117.2
```

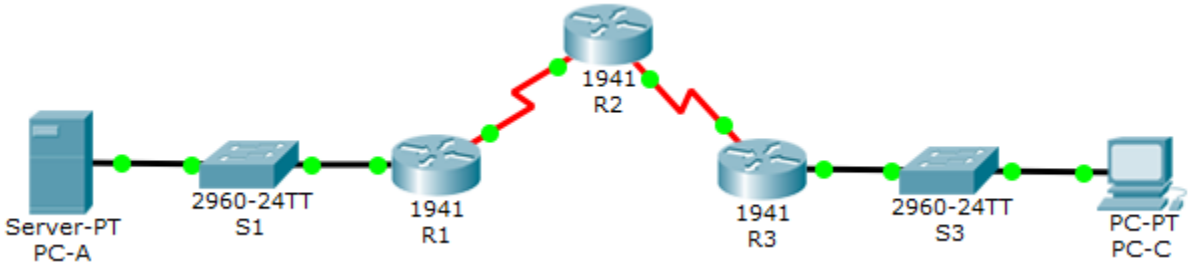
c. Exit the SSH session to SWC.
d. Ping from PCA to all of the other IP addresses in the network. If the pings are unsuccessful, verify the IP addresses before continuing.
e. SSH from PCA to SWC. The access list causes the router to reject the connection.
f. SSH from PCA to SWB. The access list is placed on G0/2 and does not affect this connection. The username is Admin, and the password is Adminpa55.
g. After logging into SWB, do not log out. SSH to SWC in privileged EXEC mode.

```
SWB# ssh -l Admin 10.101.117.2
```

Practical 12

Configuring a Zone-Based Policy Firewall (ZPF)

Topology:



Addressing Table:

Device	Interface IP	Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.1	255.255.255.252	N/A	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/6
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18

1. Verify Basic Network Connectivity

Step1: From the PC-A command prompt, ping PC-C at 192.168.3.3.

Step2: Access R2 using SSH.

- From the PC-C command prompt, SSH to the S0/0/1 interface on R2 at 10.2.2.2. Use the username Admin and password Adminpa55 to log in.

```
PC> ssh -l Admin 10.2.2.2
```

- Exit the SSH session.

Step3: From PC-C, open a web browser to the PC-A server.

- Click the Desktop tab and then click the Web Browser application. Enter the PC-A IP address 192.168.1.3 as the URL. The Packet Tracer welcome page from the web server should be displayed.
- Close the browser on PC-C.

2. Create the Firewall Zones on R3

Step1: Enable the Security Technology package.

- On R3, issue the `show version` command to view the Technology Package license information. If the Security Technology package has not been enabled, use the following command to enable the package.

```
R3(config)# license boot module c1900 technology-package
securityk9
```

- Accept the end-user license agreement.
- Save the running-config and `reload` the router to enable the security license.
- Verify that the Security Technology package has been enabled by using the `show version` command.

Step2: Create an internal zone.

- Use the zone security command to create a zone named IN-ZONE.

```
R3(config)# zone security IN-ZONE
R3(config-sec-zone)# exit
```

Step3: Create an external zone.

- Use the zone security command to create a zone named OUT-ZONE.

```
R3(config)# zone security OUT-ZONE
R3(config-sec-zone)# exit
```

3. Identify Traffic Using a Class-Map

Step 1: Create an ACL that defines internal traffic.

- Use the access-list command to create extended ACL 101 to permit all IP protocols from the 192.168.3.0/24 source network to any destination.

```
R3(config)# access-list 101 permit ip 192.168.3.0 0.0.0.255
any
```

Step 2: Create a class map referencing the internal traffic ACL.

- Use the class-map type inspect command with the match-all option to create a class map named IN-NET-CLASS-MAP. Use the match access-group command to match ACL 101.

```
R3(config)# class-map type inspect match-all
IN-NET-CLASS-MAP
R3(config-cmap)# match access-group 101
R3(config-cmap)# exit
```

4. Specify Firewall Policies

Step1: Create a policy map to determine what to do with matched traffic.

- Use the policy-map type inspect command and create a policy map named IN-2-OUT-PMAP.

```
R3(config)# policy-map type inspect IN-2-OUT-PMAP
```

Step2: Specify a class type of inspect and reference class map IN-NET-CLASS-MAP.

```
R3(config-pmap)# class type inspect IN-NET-CLASS-MAP
```

Step3: Specify the action of inspect for this policy map.

- The use of the inspect command invokes context-based access control (other options include pass and drop).

```
R3(config-pmap-c)# inspect
R3(config-pmap-c)#inspect
%No specific protocol configured in class IN-NET-CLASS-MAP for inspection. All
protocols will be inspected
```

Step4: Issue the exit command twice to leave config-pmap-c mode and return to config mode.

```
R3(config-pmap-c)# exit
R3(config-pmap)# exit
```

5. Apply Firewall Policies

Step1: Create a pair of zones.

- Using the zone-pair security command, create a zone pair named IN-2-OUT-ZPAIR. Specify the source and destination zones that were created in Task 1.

```
R3(config)# zone-pair security IN-2-OUT-ZPAIR source
IN-ZONE destination OUTZONE
```

Step2: Specify the policy map for handling the traffic between the two zones.

- Attach a policy-map and its associated actions to the zone pair using the service-policy type inspect command and reference the policy map previously created, IN-2-OUT-PMAP.

```
R3(config-sec-zone-pair)# service-policy type inspect
IN-2-OUT-PMAP
R3(config-sec-zone-pair)# exit
```

Step3: Assign interfaces to the appropriate security zones.

- Use the zone-member security command in interface configuration mode to assign G0/1 to IN-ZONE and S0/0/1 to OUT-ZONE.

```
R3(config)# interface g0/1
R3(config-if)# zone-member security IN-ZONE
R3(config-if)# exit
R3(config)# interface s0/0/1
R3(config-if)# zone-member security OUT-ZONE
R3(config-if)# exit
```

Step4: Copy the running configuration to the startup configuration.

6. Test Firewall Functionality from IN-ZONE to OUT-ZONE

Verify that internal hosts can still access external resources after configuring the ZPF.

Step1: From internal PC-C, ping the external PC-A server.

- From the PC-C command prompt, ping PC-A at 192.168.1.3. The ping should succeed.

Step2: From internal PC-C, SSH to the R2 S0/0/1 interface.

- From the PC-C command prompt, SSH to R2 at 10.2.2.2. Use the username Admin and the password Adminpa55 to access R2. The SSH session should succeed.
- While the SSH session is active, issue the command show policy-map type inspect zone-pair sessions on R3 to view established sessions.

```
R3# show policy-map type inspect zone-pair sessions
```

```
R3#show policy-map type inspect zone-pair sessions

policy exists on zp IN-2-OUT-ZPAIR
 Zone-pair: IN-2-OUT-ZPAIR

  Service-policy inspect : IN-2-OUT-PMAP

    Class-map: IN-NET-CLASS-MAP (match-all)
      Match: access-group 101
      Inspect

        Number of Established Sessions = 1
        Established Sessions
         Session 191521144 (192.168.3.3:1025)=>(10.2.2.2:22) tcp
SIS_OPEN/TCP_ESTAB
          Created 00:00:28, Last heard  00:00:22
          Bytes sent (initiator:responder) [704:804]
    Class-map: class-default (match-any)
      Match: any
      Drop (default action)
        0 packets, 0 bytes
```

- From PC-C, exit the SSH session on R2 and close the command prompt window.

Step3: From internal PC-C, open a web browser to the PC-A server web page.

- Enter the server IP address 192.168.1.3 in the browser URL field, and click `Go`. The HTTP session should succeed. While the HTTP session is active, issue the command show policy-map type inspect zone-pair sessions on R3 to view established sessions.
- Note: If the HTTP session times out before you execute the command on R3, you will have to click the `Go` button on PC-C to generate a session between PC-C and PC-A.

```
R3# show policy-map type inspect zone-pair sessions
R3#show policy-map type inspect zone-pair sessions

policy exists on zp IN-2-OUT-ZPAIR
 Zone-pair: IN-2-OUT-ZPAIR

  Service-policy inspect : IN-2-OUT-PMAP

    Class-map: IN-NET-CLASS-MAP (match-all)
      Match: access-group 101
      Inspect

        Number of Established Sessions = 1
        Established Sessions
         Session 290779304 (192.168.3.3:1027)=>(192.168.1.3:80) tcp
SIS_OPEN/TCP_ESTAB
          Created 00:00:02, Last heard  00:00:02
          Bytes sent (initiator:responder) [284:575]
    Class-map: class-default (match-any)
      Match: any
      Drop (default action)
        0 packets, 0 bytes
```

7. Test Firewall Functionality from OUT-ZONE to IN-ZONE

Verify that external hosts CANNOT access internal resources after configuring the ZPF.

Step1: From the PC-A server command prompt, ping PC-C.

- From the PC-A command prompt, ping PC-C at 192.168.3.3. The ping should fail.

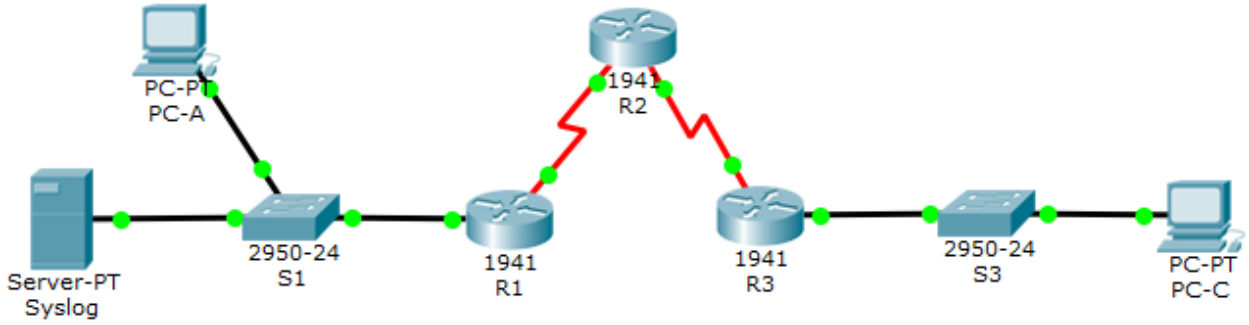
Step2: From R2, ping PC-C.

- From R2, ping PC-C at 192.168.3.3. The ping should fail.

Practical 13

Configure IOS Intrusion Prevention System (IPS) Using the CLI

Topology:



Addressing Table:

Device	Interface IP	Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/1
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/1
	S0/0/0	10.2.2.1	255.255.255.252	N/A	N/A
Syslog	NIC	192.168.1.50	255.255.255.0	192.168.1.1	S1 F0/2
PC-A	NIC	192.168.1.2	255.255.255.0	192.168.1.1	S1 F0/3
PC-C	NIC	192.168.3.2	255.255.255.0	192.168.3.1	S3 F0/2

1. Enable IOS IPS

Step1: Enable the Security Technology package.

- On R1, issue the `show version` command to view the Technology Package license information.
- If the Security Technology package has not been enabled, use the following command to enable the package.

```
license boot module c1900 technology-package securityk9
```

```
ACCEPT? [yes/no]: yes
% use 'write' command to make license boot config take effect on next boot
%IOS_LICENSE_IMAGE_APPLICATION-6-LICENSE_LEVEL: Module name = C1900 Next reboot
level = securityk9 and License = securityk9
```

- Save the running-config and reload the router to enable the security license.
- Verify that the Security Technology package has been enabled by using the show version command.

```
-----------------------------------------------------------------
Technology    Technology-package          Technology-package
              Current       Type          Next reboot
-----------------------------------------------------------------
ipbase        ipbasek9      Permanent     ipbasek9
security      securityk9    Evaluation    securityk9
data          None          None          None

Configuration register is 0x2102
```

Step2: Verify network connectivity.

- Ping from PC-C to PC-A. The ping should be successful.
- Ping from PC-A to PC-C. The ping should be successful.

Step3: Create an IOS IPS configuration directory in flash.

- On R1, create a directory in flash using the mkdir command.

```
R1# mkdir ipsdir
```

```
R1#mkdir ipsdir
Create directory filename [ipsdir]?
Created dir flash:ipsdir
```

Step4: Configure the IPS signature storage location.

```
R1(config)#ip ips config location flash:ipsdir
```

Step5: Create an IPS rule.

```
R1(config)# ip ips name iosips
```

Step6: Enable logging.

Syslog notification is enabled by default. If the logging console is enabled, IPS syslog messages display.

- Enable syslog if it is not enabled.

```
R1(config)#ip ips notify log
```

- Use the clock set command from privileged EXEC mode to reset the clock.

```
R1#clock set 08:42:00 oct 09 2024
```

- Verify that the timestamp service for logging is enabled on the router using the `show run` command. Enable the timestamp service if it is not enabled.

```
R1(config)# service timestamps log datetime msec
```

- Send log messages to the syslog server at IP address 192.168.1.50.

```
R1(config)#logging host 192.168.1.50
```

Step7: Configure IOS IPS to use the signature categories.

- Retire the all signature category with the retired true command (all signatures within the signature release). Unretire the IOS_IPS Basic category with the retired false command.

```
R1(config)#ip ips signature-category
R1(config-ips-category)#category all
R1(config-ips-category-action)#retire true
R1(config-ips-category-action)#exit
R1(config-ips-category)#category ios_ips basic
R1(config-ips-category-action)#retired false
R1(config-ips-category-action)#exit
R1(config-ips-category)#exit
```

```
R1(config-ips-category)#exit
Do you want to accept these changes? [confirm]
Applying Category configuration to signatures ...
%IPS-6-ENGINE_BUILDING: atomic-ip - 288 signatures - 6 of 13 engines
%IPS-6-ENGINE_READY: atomic-ip - build time 30 ms - packets for this engine will
be scanned
```

Step8: Apply the IPS rule to an interface.

```
R1(config)#interface GigabitEthernet0/1
R1(config-if)#ip ips iosips out
R1(config-if)#
*Oct 09, 08:54:06.5454:  %IPS-6-ENGINE_BUILDS_STARTED:  08:54:06 UTC Oct 09 2024
*Oct 09, 08:54:06.5454:  %IPS-6-ENGINE_BUILDING: atomic-ip - 3 signatures - 1 of
13 engines
*Oct 09, 08:54:06.5454:  %IPS-6-ENGINE_READY: atomic-ip - build time 8 ms -
packets for this engine will be scanned
*Oct 09, 08:54:06.5454:  %IPS-6-ALL_ENGINE_BUILDS_COMPLETE: elapsed time 8 ms
```

2. Modify the Signature

Step1: Change the event-action of a signature.

- Un-retire the echo request signature (signature 2004, subsig ID 0), enable it, and change the signature action to alert and drop.

```
R1(config)#ip ips signature-definition
R1(config-sigdef)#signature 2004 0
R1(config-sigdef-sig)#status
R1(config-sigdef-sig-status)#retired false
R1(config-sigdef-sig-status)#enabled true
R1(config-sigdef-sig-status)#exit
R1(config-sigdef-sig)#engine
```

```
R1(config-sigdef-sig-engine)#event-action produce-alert
R1(config-sigdef-sig-engine)#event-action
deny-packet-inline
R1(config-sigdef-sig-engine)#exit
R1(config-sigdef-sig)#exit
R1(config-sigdef)#exit
```

```
R1(config-sigdef)#exit
Do you want to accept these changes? [confirm]
%IPS-6-ENGINE_BUILDS_STARTED:
%IPS-6-ENGINE_BUILDING: atomic-ip - 303 signatures - 3 of 13 engines
%IPS-6-ENGINE_READY: atomic-ip - build time 480 ms - packets for this engine will
be scanned
%IPS-6-ALL_ENGINE_BUILDS_COMPLETE: elapsed time 648 ms
```

Step2: Use show commands to verify IPS.

- Use the `` `show ip ips all` `` command to view the IPS configuration status summary.

Step3: Verify that IPS is working properly.

- From PC-A, attempt to ping PC-C. The pings should fail because the IPS rule for event-action of an echo request was set to "deny-packet-inline".
- From PC-A, attempt to ping PC-C. The ping should be successful. This is because the IPS rule does not cover echo reply. When PC-A pings PC-C, PC-C responds with an echo reply.

Step4: View the syslog messages.

